PERTEMUAN 9

INTEGRAL LIPAT

Definisi Integral Lipat Dua:

Misalkan f suatu fungsi dua peubah yang terdefinisi pada suatu persegi panjang tertutup *R*. Jika $\lim_{|P|\to 0} \sum_{k=1}^{n} f(\bar{x}_k, \bar{y}_k)\Delta A_k$ ada, kita katakan f dapat diintegralkan pada *R*. Lebih lanjut $\iint_R f(x,y)\,dA = \iint_R f(x,y)\,dx\,dy$ yang disebut integral lipat dua f pada *R* diberikan oleh:

$$\iint_R f(x,y)\,dA = \lim_{|p|\to 0} \sum_{k=1}^{n} f(\bar{x}_k, \bar{y}_k)\Delta A_k$$

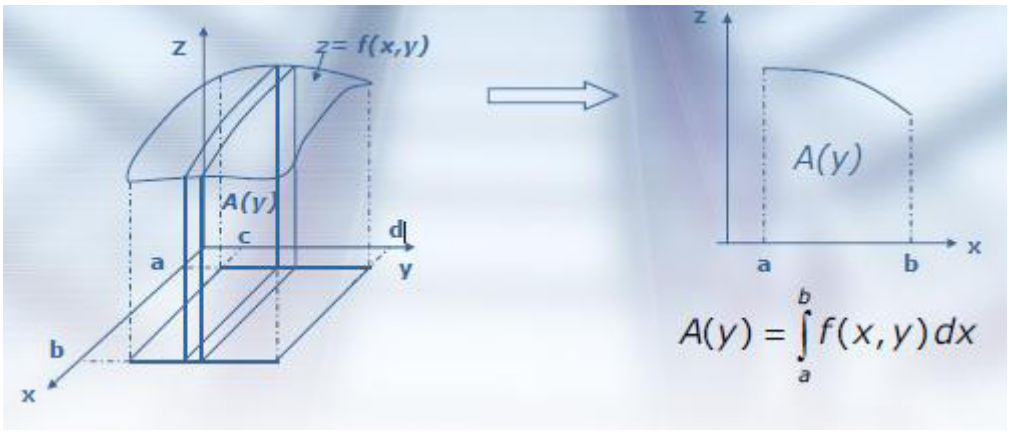
atau

$$\iint_R f(x,y)\,dx\,dy = \lim_{|p|\to 0} \sum_{k=1}^{n} f(\bar{x}_k, \bar{y}_k)\Delta x_k\,\Delta y_k$$

Jika $z = f(x,y)$ kontinu, $f(x,y) \geq 0$ pada persegi panjang *R*, maka di bawah permukaan $z = f(x,y)$ dan di atas *R*.

Jika $f(x,y) \geq 0$ pada *R*, maka volume dapat dihitung dengan metode irisan sejajar, yaitu:

1) Sejajar bidang *XOZ*

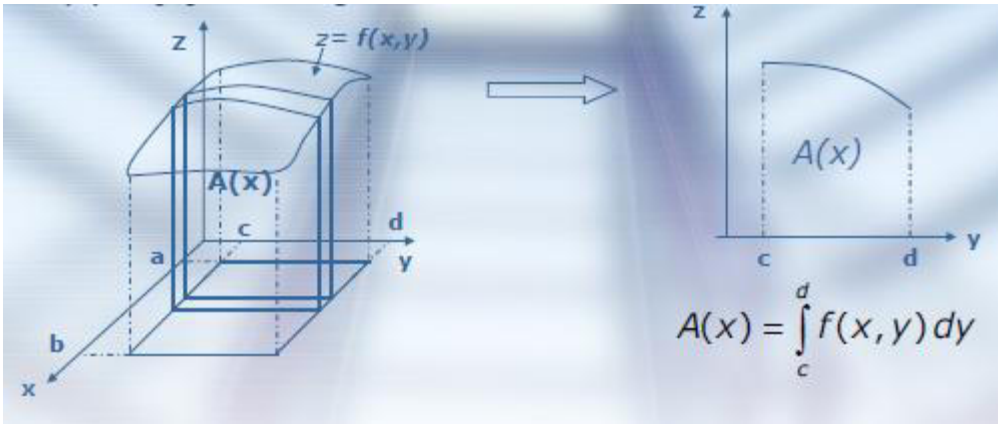


$$\iint_R f(x,y)\,dA = \int_c^d A(y)\,dy = \int_c^d \left[\int_a^b f(x,y)\,dx\right] dy = \int_c^d \int_a^b f(x,y)\,dx\,dy$$

Maka

$$\iint_R f(x,y)\,dA = \int_c^d \int_a^b f(x,y)\,dx\,dy$$

2) Sejajar bidang *YOZ*



$$\iint_R f(x,y)\,dA = \int_a^b A(y)\,dy = \int_a^b \left[\int_c^d f(x,y)\,dx\right] dy = \int_a^b \int_c^d f(x,y)\,dx\,dy$$

Maka

$$\iint_R f(x,y)\,dA = \int_a^b \int_c^d f(x,y)\,dx\,dy$$

Contoh Soal:

1. Hitung $\int_0^3 \left[\int_1^2 (2x+3y)dx\right] dy$!

   Jawab:

   Pada integral sebelah dalam *y* berupa konstanta, sehingga:

   $$\int_1^2 (2x+3y)dx = [x^2+3yx]_1^2 = 4+6y-(1+3y) = 3+3y$$

   Akibatnya,

   $$\int_0^3 \left[\int_1^2 (2x+3y)dx\right] dy = \int_0^3 [3+3y]\,dy = \left[3y+\frac{3}{2}y^2\right]_0^3 = 9+\frac{27}{2} = \frac{45}{2}$$

2. Hitung $\int_1^2 \left[\int_0^3 (2x+3y)dy\right] dx$!

   Jawab:

   Pada integral sebelah dalam *y* berupa konstanta, sehingga:

   $$\int_0^3 (2x+3y)dy = \left[2xy+\frac{3}{2}y^2\right]_0^3 = 6x+\frac{27}{2}$$

   Akibatnya,

   $$\int_1^2 \left[\int_0^3 (2x+3y)dy\right] dx = \int_1^2 \left[6x+\frac{27}{2}\right] dx = \left[3x^2+\frac{27}{2}x\right]_1^2 = 12+27-\left(3+\frac{27}{2}\right) = \frac{45}{2}$$

3. Hitung $\int_0^8 \int_0^4 \frac{1}{16}(64 - 8x + y^2)\, dx\, dy$!

   Jawab:

$$\int_0^4 \frac{1}{16}(64 - 8x + y^2)\, dx = \frac{1}{16}[64x - 4x^2 + y^2 x]_0^4$$

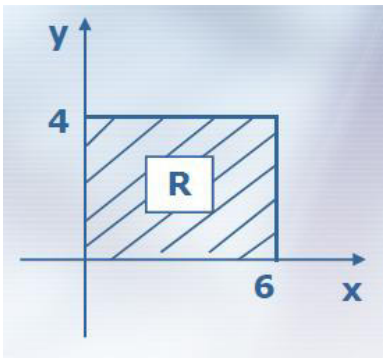
$$= \frac{1}{16}[256 - 64 + 4y^2]$$

$$= 12 + \frac{1}{4}y^2$$

$$\int_0^8 \left(12 + \frac{1}{4}y^2\right) dy = \left[12y + \frac{y^3}{12}\right]_0^8 = 96 + \frac{512}{12} = 96 + \frac{128}{3} = 138\frac{2}{3}$$

4. Hitung integral lipat dua berikut ini: $\iint_R x^2 + 2y^2\, dA$ dimana $R = \{(x, y) | 0 \le x \le 6, 0 \le y \le 4\}$

   Jawab:



$$\iint_R (x^2 + 2y^2)\, dA = \int_0^6 \int_0^4 (x^2 + 2y^2)\, dy\, dx$$

$$= \int_0^6 \left[x^2 y + \frac{2}{3}y^3\right]_0^4 dx$$

$$= \int_0^6 \left(4x^2 + \frac{128}{3}\right) dx$$

$$= \left[\frac{4}{3}x^3 + \frac{128}{3}x\right]_0^6$$

$$= 288 + 256$$

$$= 544$$

atau

$$\iint_R x^2 + 2y^2\, dA = \int_0^4 \int_0^6 x^2 + 2y^2\, dx\, dy$$

$$= \int_0^4 \left[\frac{1}{3}x^3 + 2xy^2\right]_0^6 dy$$
$$= \int_0^4 (72 + 12y^2) dy$$
$$= [72x + 4x^3]_0^4$$
$$= 288 + 256$$
$$= 544$$

5. Hitunglah integral lipat dua berikut ini: $\iint_R \sin(x+y)\, dA$ dimana $R = \left\{(x,y)|0 \leq x \leq \frac{\pi}{2},\ 0 \leq y \leq \frac{\pi}{2}\right\}$!

   Jawab:

$$\iint_R \sin(x+y)\, dA = \int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{2}} \sin(x+y)\, dy\, dx$$
$$= \int_0^{\frac{\pi}{2}} [-\cos(x+y)]_0^{\frac{\pi}{2}} dx$$
$$= \int_0^{\frac{\pi}{2}} \left(-\cos\left(\frac{\pi}{2}+y\right) + \cos y\right) dx$$
$$= [\sin y]_0^{\frac{\pi}{2}} - \left[\sin\left(\frac{\pi}{2}+y\right)\right]_0^{\frac{\pi}{2}}$$
$$= \sin\left(\frac{\pi}{2}\right) - \sin(\pi) + \sin\left(\frac{\pi}{2}\right)$$
$$= 2$$

Latihan Soal

1. Hitunglah integral lipat di bawah ini!

a.	$\int_0^2 \int_1^3 x^2 y\ dy\, dx$	f.	$\int_0^{\frac{\pi}{2}} \int_0^1 x \sin xy\ dy\, dx$
b.	$\int_{-1}^4 \int_1^2 (x + y^2)\ dy\, dx$	g.	$\int_0^1 \int_0^1 xe^{xy}\ dy\, dx$
c.	$\int_1^2 \int_0^3 (xy + y^2)\ dy\, dx$	h.	$\int_0^3 \int_0^1 2x\sqrt{x^2+y}\ dx\, dy$

d.	$\int_{-1}^{1}\int_{1}^{2}(x^2+y^2)\ dx\ dy$	i.	$\int_{0}^{1}\int_{0}^{1}\frac{y}{(xy+1)^2}\ dx\ dy$
e.	$\int_{0}^{\pi}\int_{0}^{1}x\sin y\ dx\ dy$	j.	$\int_{0}^{1}\int_{0}^{2}\frac{y}{1+x^2}\ dy\ dx$

2. Hitunglah nilai dari $\iint_R f(x,y)\,dx\,dy$ untuk fungsi:

a.	$f(x,y)=(x+2y)^2$ dengan $R=[-1,2]\times[0,2]$
b.	$f(x,y)=xy^3$ dengan $R=[-1,1]\times[0,2]$
c.	$f(x,y)=x^2+y^2$ dengan $R=[0,1]\times[0,1]$
d.	$f(x,y)=\sin(x+y)$ dengan $R=\left[0,\frac{\pi}{2}\right]\times\left[0,\frac{\pi}{2}\right]$
e.	$f(x,y)=y^2\cos^2 x$ dengan $R=\left[-\frac{\pi}{2},\pi\right]\times[1,2]$